

Pavankumar Chandankar

Applied AI & Machine Learning | Scientific Research | Data Analytics | Interdisciplinary Systems

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PROFILE

- M.Sc. candidate in Mechatronics at the University of Siegen with research experience in applied artificial intelligence and machine learning, including PyTorch model development, empirical evaluation, large-scale experimentation, and scientific publication. Student Assistant in Ubiquitous Computing Department and first author of two peer-reviewed papers, including the iWOAR 2026 Best Paper. Previous embedded systems engineer with experience spanning project management, control systems, machine learning, user-centered AI design, and industrial engineering environments in Germany and India.

AWARDS & ACHIEVEMENTS

- Best Paper Award** ■ **iWOAR 2026 Best Paper Award** for “Integrating Adaptive Sampling into Deep Learning-Based Human Activity Recognition.”
- Gold Medalist** ■ University rank 1 in B.E. Mechanical Engineering with a final grade of **1.3**.

PUBLICATIONS

- 2026** ■ **Pavankumar Chandankar**, Marius Bock, Juergen Gall, Michael Moeller, and Kristof Van Laerhoven.
“Integrating Adaptive Sampling into Deep Learning-Based Human Activity Recognition.”
iWOAR 2026, Springer LNCS; **Best Paper Award, iWOAR 2026**; full paper accepted, publication forthcoming.
- 2025** ■ **Pavankumar Chandankar** and Robin Burchard.
“Sensor-Specific Transformer (PatchTST) Ensembles with Test-Matched Augmentation.”
UbiComp Companion, ISWC 2025, ACM; **DOI: 10.1145/3714394.3756196**

RESEARCH EXPERIENCE

- Oct 2025 – Present** ■ **Student Assistant** *Ubiquitous Computing Department, University of Siegen, Germany*
- Designed and empirically evaluated PyTorch-based machine-learning approaches for adaptive and resource-efficient human activity recognition using wearable sensor data.
 - Planned reproducible experiments across datasets and sampling conditions, managed large-scale workflows on the OMNI HPC cluster, and contributed results to peer-reviewed publications and conference presentations.

TECHNICAL & RESEARCH SKILLS

- Applied AI & Data** ■ Python, SQL, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, time-series analysis, classification, anomaly detection, model evaluation, and model compression.

TECHNICAL & RESEARCH SKILLS (continued)

- Research & UCD** ■ Scientific literature review, experiment design, User-Centered Design, semi-structured interviews, affinity mapping, scenario development, user personas, usability evaluation, scientific writing, and publication preparation.
- Tools & Infrastructure** ■ Git, SLURM/HPC, MLflow, AWS SageMaker, experiment tracking, Matplotlib, PyTest, CI/CD, and LaTeX.
- Engineering Systems** ■ C++ (Basics), CAN bus, MQTT, ROS, dSPACE HIL, Vector CANoe, microcontrollers, PLCs, and sensor calibration.

INDUSTRY EXPERIENCE

- Oct 2022 – Oct 2023** ■ **GET- Embedded Systems Engineer** *Larsen & Toubro Technology Services, India*
- Designed and validated CAN-bus communication for railway control systems using dSPACE HIL and Vector CANoe, identifying integration issues before deployment.
 - Automated Python-based testing and code-coverage analysis within CI/CD workflows and supported project planning and delivery using Oracle Primavera P6.
- Sep 2021 – Feb 2022** ■ **Process Engineering Intern** *Tata Hitachi, India*
- Worked with microcontrollers, robotic control systems, PWM, and Mitsubishi PLCs to simulate material-handling systems and diagnose control-level faults.

SELECTED PROJECTS

- Workplace AI** ■ MayDay – Workplace AI Companion. Co-developed a user-centered AI workplace companion for managing interruptions, focus, and productivity. Applied semi-structured interviews, affinity mapping, scenario-based design, user personas, interactive wireframes, and user evaluation to iteratively refine the system. **DOI:** [10.13140/RG.2.2.13069.65763](#)
- HAR** ■ Human Activity Recognition using Neural Networks. Multimodal activity recognition using DTW, DBA, TCN, and LSTM models. [GitHub](#)
- CAN Bus** ■ Safety-Critical Automotive CAN Bus Anomaly Detection. LSTM-based anomaly detection for automotive CAN-network data. [GitHub](#)

EDUCATION

- Oct 2023 – Present** ■ **M.Sc. Mechatronics** *Universität Siegen, Germany*
- Master's thesis:** Weight Transfer for Adversarially Robust Neural Networks via Weight-Space Learning
- Aug 2018 – Jul 2022** ■ **B.E. Mechanical Engineering** *Visvesvaraya Technological University, India*
- Graduation grade:** 9.23/10; **German equivalent:** 1.3

LANGUAGES

- Kannada** ■ Native speaker.
- English** ■ C1; IELTS; academic and professional working language.
- German** ■ A1 completed; A2 currently in progress.